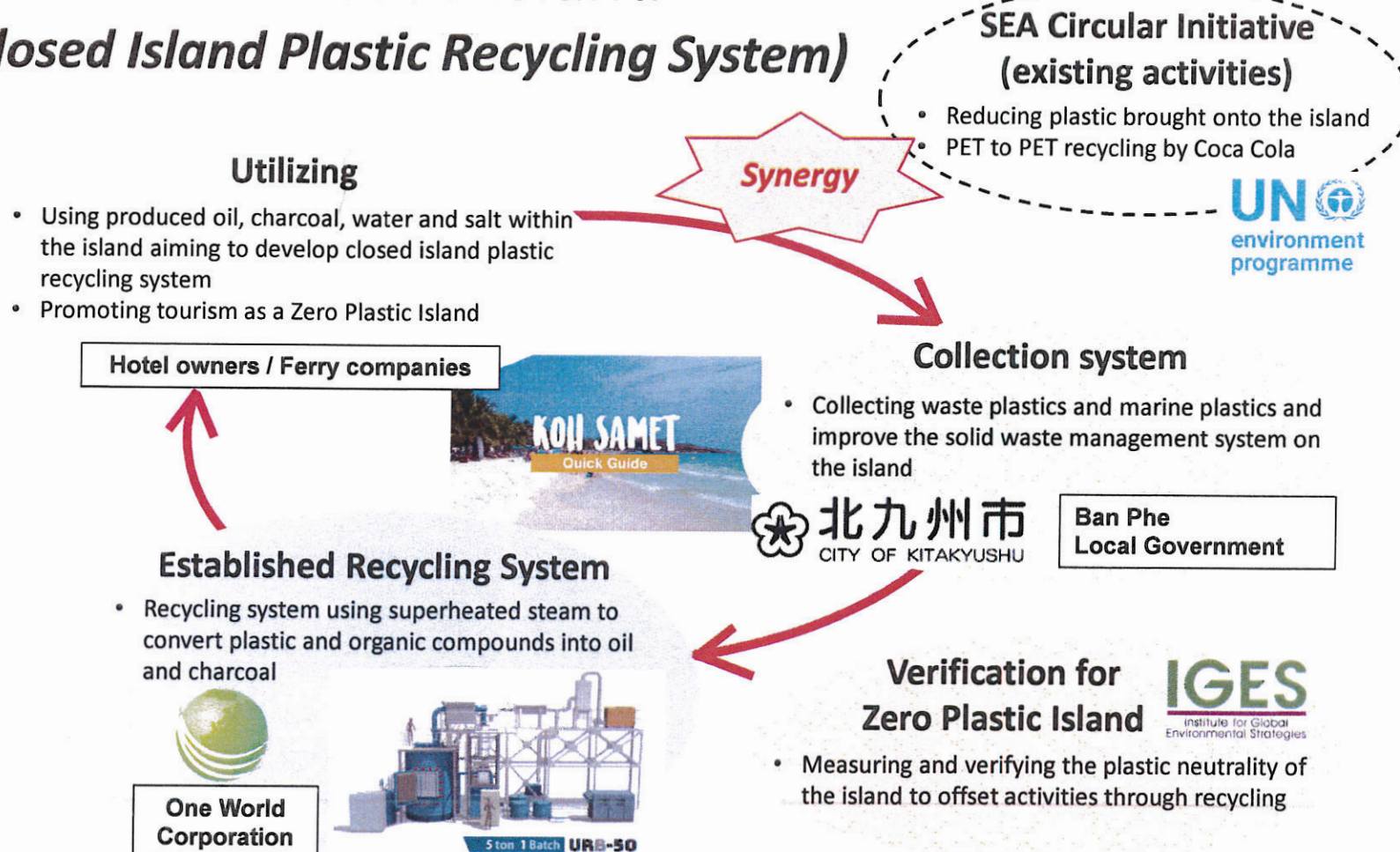


Zero Plastic Island

(Closed Island Plastic Recycling System)



Summary -- Project Code -- Zero Plastic Island (Development of a Closed Island Plastic Recycling System)

Summary of Project / Partnership		
- This project aims to establish a closed island recycling system for plastic waste generated in Phe district, Rayong Province, Thailand using the "URBAN RIG" (a recycling system using superheated steam change any types of plastics and organic in oil and charcoal in respectively) developed by One World Corporation. The project aims to use the oil recovered by URBAN RIG as a heat source to produce power for the independent operation of the plant (excess power will be used for sorting equipment in operation at the site) and to generate concentrated seawater (salt) as a byproduct of the use of seawater for heat exchange in condensers. Besides, the recovered charcoal will be used as a heat source for barbeques at resort hotels on the island. - Local counterparts will include the Ministry of Natural Resources and Environment of Thailand and Phe district (Samet island), with the City of Kitakyushu and Institute for Global Environmental Strategies (IGES) participating as partner organisations from Japan. Kitakyushu has already implemented a support project to improve local waste management including a promotion of environmental education, and this project will continue to support the establishment of a waste sorting and collection model. Kitakyushu has also concluded an agreement on the development of a joint framework for plastic waste management with the UNEP Regional Office for Asia and the Pacific to implement a collection programme for plastic bottles in Phe district (Samet island), which aims to achieve a synergy with this initiative. - The proposed project will cover all four Alliance pillars: (1) the development of innovative technology and (2) infrastructure for collecting and recycling waste plastics and marine plastics with (3) the promotion of environmental education and engagement of multi-stakeholders, and (4) beach and ocean clean-up activities for marine plastics.		
Alignment with AEPW Selection Criteria		
License to operate	H	Local permit and licensing systems must be confirmed.
Strategic Alignment	H	The local government of Phe district in Rayong Province, Thailand where is a planned project site will become the project owner. The proposed project will cover all four Strategic Alignments of AEPW.
Technical feasibility	H	One World's URBAN RIG is in operation in four locations in Japan, with a proven level of technical feasibility.
Economics	H	Profitability must be considered to ensure economic efficiency (An expected income and expenditure statement will be attached to a proposal).
Plastic Waste Impact	H	This project will create a Zero Plastic Island* for the entire Samet island of Phe district, treating about 3,300 tonnes of plastic waste annually.
Stakeholder Engagement & Credibility	H	Kitakyushu City is implementing a support project with a local government of Phe district to improve waste management utilizing a collaborative framework with the UNEP Regional Office for Asia and the Pacific and the city-to-city collaboration programme, which has already established a relationship of trust with the local C/P.
Environmental & Social Impact	H	From an environmental perspective, this project will reduce the amount of waste sent to landfill and improve the environment at the dump site on the island and can also be utilised as a source of backup power in the event of a disaster (e.g. blackout). From a social perspective, the development of a model island free from marine plastic will add value to the resort island and help promote tourism, creating local employment opportunities and increasing the income of local residents.
Objectives & Contribution / Milestones		
- The One World's resource recovery plant (URBAN RIG) will treat marine plastic and other waste plastic on the island and establish a recycling system using the produced oil, charcoal, water and salt, creating an Zero Plastic Island*. This will help promote tourism as a resort island. - In the first year of implementation, an installation plan (site selection, plant design, acquisition of permits and licensing, etc.) will be carried out for the resource recovery plant (URBAN RIG), a waste separation and collection system will be implemented and methods for the use of recycled resources (heat, oil, charcoal, water, salt, etc.) will be decided. In addition, a 1-tonne URBAN RIG plant will be installed on-site for a demonstration on operation and management. The second year will include installation work and a trial operation of the plant, with the third year targeting the completion of recycling infrastructure on the entire island and verification of the Zero Plastic Island project's objectives.		
Expected Alliance Engagement and Funding		
- URBAN RIG is an innovative plant that can produce recovered oil from marine plastic and other waste plastic. One feature of the plant is that it can treat not only waste plastics including marine plastic, but other types of waste as well and is capable of flexibly responding to situations at installation sites. Since the island has often no waste treatment infrastructure and is facing a problem with marine plastics, this resource recovery plant will be installed on-site to create a closed recycling system on the island. - A support of providing a project budget (USD 4.3 million) as well as advices for the development of a closed island plastic recycling system is expected from AEPW.		
Other Observations		
- One World's URBAN RIG will be used to establish a closed island recycling system for recovering resources from marine plastic and other waste plastic on the island. This type of mechanism will be used to build a sustainable system that can be expanded to other regions. - Specifically, a mechanism will be developed in the first year, which can be used to offset activities by eliminating plastic waste in the environment, and credits (tentative name: Marine Plastic Credit) will be created according to the amount of plastic collected and treated once the plant is in operation in the third year. This will be applied to the aim to create an island model that will allow the entire island to become a neutral for a leakage of plastic waste in the environment (*referred to as "Zero Plastic Island") and help promote local tourism as a resort island.		
Recommendation : Pass Gate 1		

Feedstock sourcing

Field survey

- Joint field survey was conducted in August, 2019 (UNEP-ROAP, City of Kitakyushu, Phe sub-district administration organization, etc.)
- Amount of solid waste collected by three trucks within the island is **10 ton/day** (from hotels and households)
- A new sorting center was established in June 2019.
- Accumulated waste at a dump was all removed and cleaned-up in January 2020.
- City of Kitakyushu will provide technical support for source separation and efficient collection of solid waste (including composting).



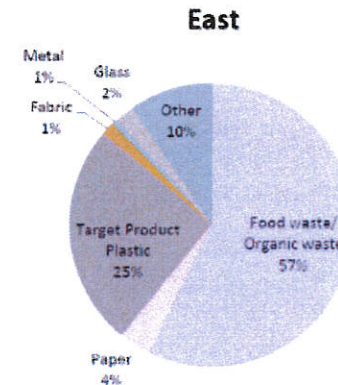
Sorting center in Samet island



Dump site in Samet island

Waste Composition Analysis

- UNEP-ROAP together with the Petroleum and Petrochemical College (PPC) research team conducted detailed waste composition analysis including Phe sub-district.
- The composition of plastic waste is **25%**.

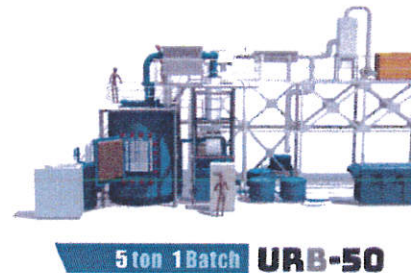


Source: the Petroleum and Petrochemical College (PPC)

Feedstock to URBAN RIG

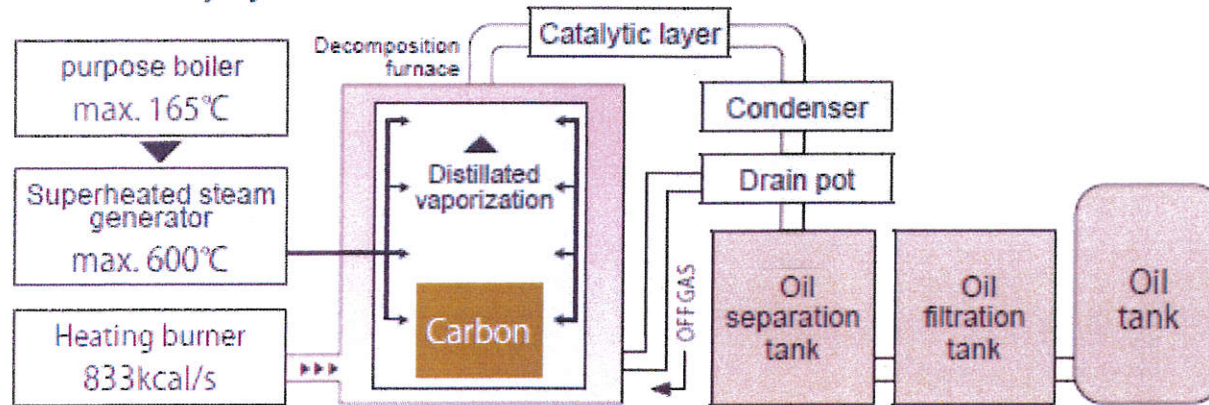
$$10 \text{ t/d} \times 25\% \div 0.25 \text{ t/m}^3 = 10 \text{ m}^3 \text{ /d}$$

Amount of waste collected Plastic contents Bulk density of plastics

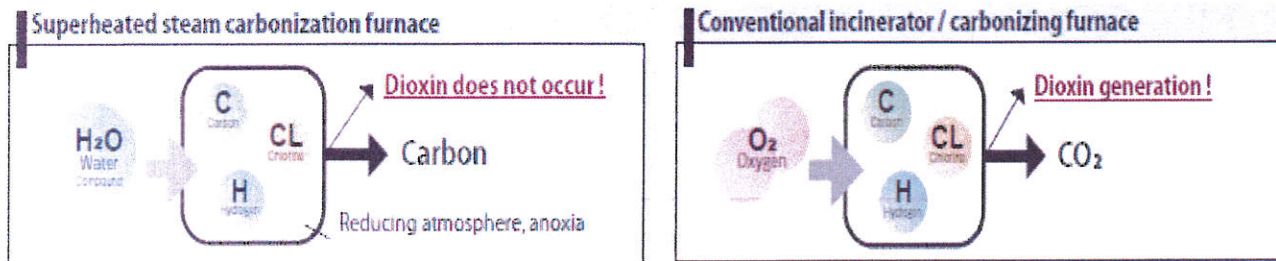


URBAN RIG – mechanism -

Oil recovery system



How is the URBAN RIG different from an incineration?

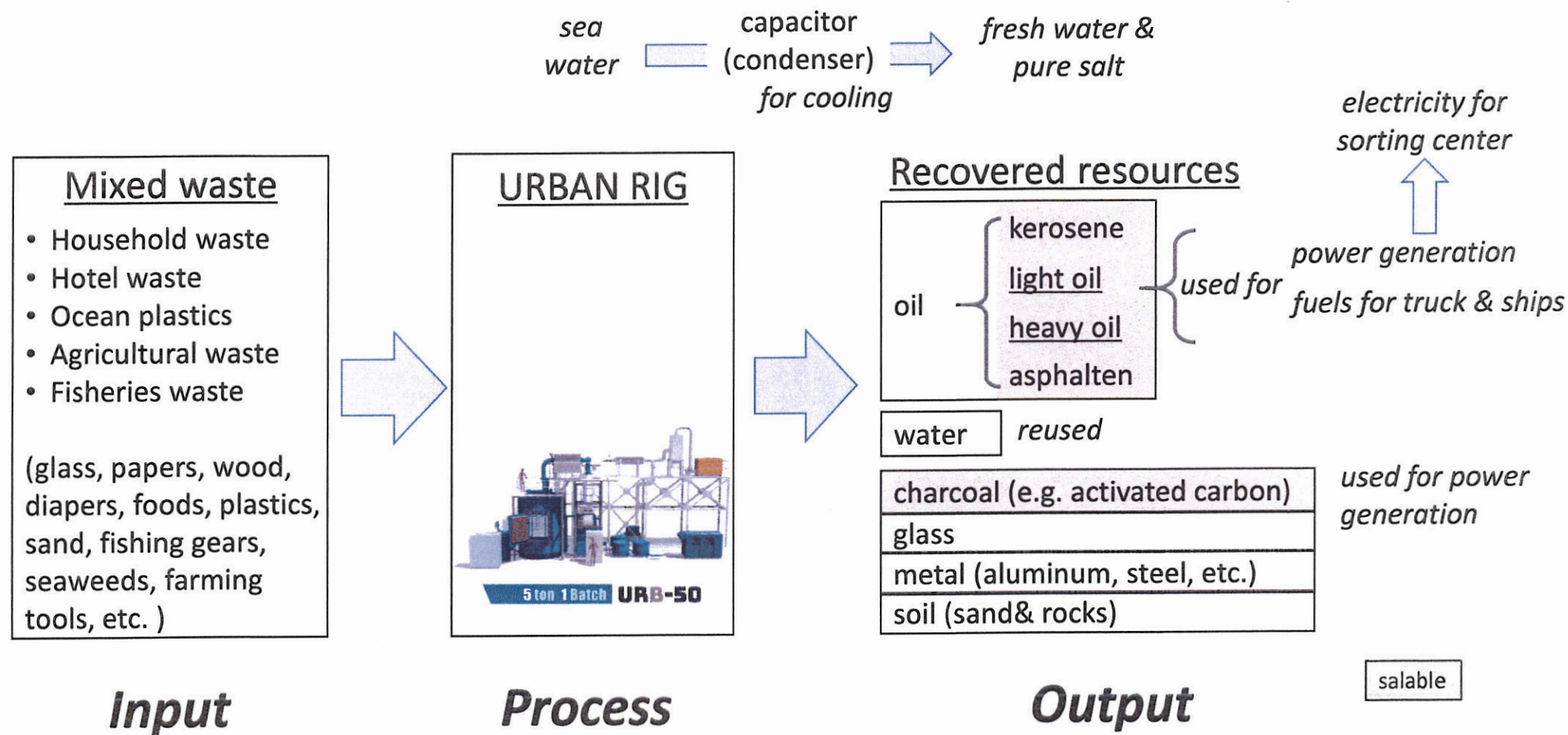


Characteristics of URBAN RIG

1. URBAN RIG can **make a profit** by recovering salable resources (carbide, light oil, mixed oil, off-gas). Non-carbide materials (e.g. glass, metals, etc.) are recyclables.
2. URBAN RIG can treat a **mixed waste of various types of waste** including organics, plastics, glass, metals, papers, woods, etc.
3. URBAN RIG can treat **ocean plastics and PVC** with corrosive chemicals (e.g. NaCl, HCL).
4. URBAN RIG can create **pure water from sea water** by using a heat generated by the equipment.
5. Light oil generated by URBAN RIG can be used for **generating electricity** as well as fuels for cars & trucks.

*URBAN RIG can treat waste and provide **self-sufficient infrastructure in an island** (e.g. water, electricity, oil, charcoal)

URBAN RIG – mass balance -



Marine Plastic Credit

1st Step: Cost-effectiveness indicator (project-based)

- Development of **monitoring system** (e.g. indicators) for measuring the amount of plastic waste recycling or reduced.
- A **reporting format** sharing to investors (e.g. AEPW) will be developed along with a global discussion on corporate disclosure on plastic-related information (e.g. PRI's Plastic Investor Working Group)
- Measuring the amount of plastic waste reduction will lead to an evaluation **criteria for ESG investment**.



An overview of the global plastics supply chain



Anthesis

2nd step: Recognition of plastic neutrality (within a regional boundary)

- **Marine Plastic Credit** will be created according to the amount of plastic collected and treated. (e.g. Marine Plastic Credit: ton of plastics input to the recycling process — residues come out from the process)
- Measuring and verifying the **plastic neutrality** of the island by the offset activities
- The island will **be recognized as a neutral** for a leakage of plastic waste in the environment (e.g. Zero Plastic Island) under the framework of **UNEP's SEA Circular Initiative**.



3rd step: Tradable crediting mechanism (in a cooperation with other organizations)

- Tradable mechanism will be developed in a cooperation with other organizations
- A mechanism of offset credit registration, verification and certification, issuance, invalidation, etc. referring to the existing carbon credit mechanism.



Economics (profit & loss)

Total input 10,000 kg

Contents	quantity/day		carbide&residues/day		quantity/day			
	composition	input	residual rate	carbide&residues	extraction rate	extraction quantity	extraction volume (bulk density: 0.8)	
Plastic	24.5 %	2,450 kg	1.3 %	31.85 kg	80 %	1960 kg	2450	
Paper	6.2 %	620 kg	8 %	49.60 kg	0 %	0 kg	0	
Fiver	1.2 %	120 kg	10 %	12.00 kg	70 %	84 kg	105	
Organic	53.8 %	5,380 kg	9 %	484.20 kg	0 %	0 kg	0	
Glass	5.7 %	570 kg	100 %	570.00 kg	0 %	0 kg	0	
Metal	0.5 %	50 kg	100 %	50.00 kg	0 %	0 kg	0	
Wood, bamboo	0.2 %	20 kg	14 %	2.8 kg	0 %	0 kg	0	
Other	7.9 %	790 kg	16 %	126.4 kg	0 %	0 kg	0	
		10,000 kg		1326.85 kg		2,044 kg	2555	

Economic model of URBAN RIG

- CAPEX: 4m USD (2m USD × 2)
- OPEX: self-sustained for fuel consumption
- Annual sales of recovered resources (oil & charcoal): 320,000USD/year
- Additional consideration of reduction of waste shipment & treatment costs (additional benefits)

Recovered resources		ratio	quantity/day	Market price	Income			
					Day		Month (30 days)	
carbonaceous residues	carbide (charcoal)	7.07 %	706.9 kg	¢ 1 /kg	7	USD	212	USD
Oil	light oil	24.5 %	2450 ℓ	¢ 35 /ℓ	858	USD	25,725	USD
	mixed oil	1.05 %	105 ℓ	¢ 20 /ℓ	21	USD	630	USD
	off-gas	3.32 %	597870 ℓ					
non-carbide	glass	5.7 %	570 kg					
	metals	0.5 %	50 kg					
water	water	47.92 %	4792 ℓ					
	evaporation	9.94 %	994 ℓ					
					886	USD	26,567	USD

Amount of self-consumption is 180L/day. The rest (2,270L/day) is used for generating electricity.

Pure water is produced from sea water by utilizing a heat generated at condenser.

* Exchange rate: assume 1USD = 100 JPY

1. I understand that a basic sorting center for 10tpd has been commissioned in June 2019. Could you:
 - a. Confirm that the Island is generating ~10tpd of total waste ?
 - b. Provide current sorting center mass balance (input and output, valorization and destination of recyclates and residuals) and how such a mass balance would look like after the project implementation ?
2. Urban RIG module from One World Corporation seems to be able to handle all kinds of plastic waste.
 - a. Could you provide track records of operations in Japan ? Since when in operations ? What learnings ? What type of maintenance is planned ?
 - b. Do you have Life Cycle Analysis (LCA) data for it ? How does it compare to alternative like sanitized landfilled or incineration ?
3. Economics. Could you provide a P&L with EBITDA, EBIT and SEGE (ie your estimation of tax, depreciation, amortization,...)
 - a. Main revenue will come from selling light oil at 35cents/L. Can you confirm this level of price is what can be achieved at Samet Island ? Who are the considered buyers ? Have they been engaged ? Is the quality/specification of the light oil suitable and understood by offtakers ? Have you assessed regulations governing sales of fuel at that location ?
 - b. I have attached a simple P&L model. Would be great if you could populate the "input" cells. I'm keen to understand the "bankability" of such a system and therefore its replicability
 - c. Please confirm that Urban RIG is license free
4. Operation. Could you clarify which entity will own the Urban RIG module and operate it ?
5. Marine plastic credits. I'd like to understand more in details your proposal, in particular step 3. Also how would this impact the project economics ?

1. I understand the a basic sorting center for 10tpd has been commissioned in June 2019. Could you:
 - a. Confirm that the Island is generating ~10tpd of total waste ?
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Total waste generating in the island:

- 10 tpd is the amount of collected by three trucks (one operated by the public and two operated by the private)
- Some of the solid waste is carrying into the dump site by their own
- Cans, bins, and PET bottles are carried in the collection boxes (PET bottles are sent to PTT Global Chemical Public Company Limited & SGC Chemicals Co., Ltd. under UNEP's SEA Circular Initiative)
- Uncollected waste

The sorting center established in June 2019:

- The sorting center is not connected with the grid power system, and so it is not in full operation.
- The information of mass balance is currently limited.

We would like to check the detail mass balance when we will do a demonstration.

2. Urban RIG module from One World Corporation seems to be able to handle all kind of plastic waste.
 - a. Could you provide track records of operations in Japan ? Since when in operations ? What learnings ? What type of maintenance is planned ?
 - b. Do you have Life Cycle Analysis (LCA) data for it ? How does it compare to alternative like sanitized landfilled or incineration ?

Track records of operations in Japan:

No.	Place	Capacity	Term of operation
1	Toyooka, Hyogo prefecture	5t/d	2 years and 3 months
2	Fukuroi, Shizuoka prefecture	5t/d	1 year and 3 months
3	Fuji, Shizuoka prefecture	5t/d	6 months
4	Setouchi, Okayama prefecture	5t/d	will be in operation from August 2020
5	Tokoname, Aichi prefecture	5t/d	will be in operation from August 2020
6	Okayama, Okayama prefecture	1t/d	8 months
7	Okinawa prefecture	1t/d	9 months



- received UMIGOMI ZERO Award (award for no marine plastic litter) in 2019 from Ministry of the Environment, Japan and the Nippon Foundation (https://uminohi.jp/umigomizero_award2020/announcement.html)
- It has been increasing our business performance while our past work performance as well as name recognition are not enough.
- For maintenance, we will inspect a thickness of furnace wall, a heating burner, a boiler etc. Periodic inspection (every a couple of months) is charged for 2% of CAPEX by dispatching a staff from factories manufacturing URBAN RIG (four places in Japan).

LCA for URBAN RIG in a comparison with LF & Incineration:

- Currently, no LCA data is available for URBAN RIG, but we will try to provide a LCA result as necessary.

3. Economics. Could you provide a P&L with EBITDA, EBIT and SEGE (ie your estimation of tax, depreciation ,amortization,...)
- Main revenue will come from selling light oil at 35cents/L. Can you confirm this level of price is what can be achieved at Samet Island ? Who are the considered buyers ? Have they been engaged ? Is the quality/specification of the light oil suitable and understood by offtakers ? Have you assess regulations governing sales of fuel at that location ?
 - I have attached a simple P&L model. Would be great if you could populate the "input" cells. I'm keen to understand the "bankability" of such a system and therefore it's replicability
 - Please confirm that Urban RIG is license free

Revenue:

- The selling price of 35 cents/L is based on the performance in Japan. We would like to check the level of price of selling light oil when we will conduct a demonstration with 1t/d URBAN RIG.
- Buyers of light oil is waste oil dealing companies and we sell our light oil as good quality waste oil (thus, we do not hold a license to sell light oil) which meets their required quality and quantity.
In a demonstration in Samet island, we would like to look for buyers of our waste oil including related regulations of sales of fuels.

Bankability :

- Please find the attached excel sheet. Let me note that the calculation does not include additional consideration of sales of charcoal, mixed oil, and other resources (e.g. metal, glass) as well as a benefits of reduction of waste treatment costs and shipping costs, which greatly contribute to the bankability of the economic model

License :

- No license for using URBN RIG
- License contract is signed with three companies in China and currently being adjusted with a company in Indonesia. In China 50 URBAN RIG (1t/d) is manufacturing for treating medical wastes under the license contract.

4. Operation. Could you clarify which entity will own the Urban RIG module and Operate it ?

Owner of URBAN RIG:

- So far, we suppose that the owner and operator of the URBAN RIG in Samet Island would be Rayong Provincial government and Phe sub-district administration organization respectively.
We will further discuss this matter with Rayong Provincial government and Phe sub-district administration organization to make an agreement.

5. Marine plastic credits. I'd like to understand more in details your proposal, in particular step 3. Also how would this impact the project economics ?

Marine Plastic Credit:

- While we will consider the possibility of creating tradable crediting mechanism, we do not aim to achieve the step 3 within the project term (we target to achieve step 2 which is a recognition of plastic neutrality.)
Therefore, the credits created by implementing the project does not impact the project economics.